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Research Project

**Optimizing Applications of Deep Learning Models for Genomics Problems to the
Subject Area of Secondary Structures in Stem Cells**

Submitted by Nesterenko Alisa Vadimovna
student of group #215, 4th year of study

Approved by the Supervisor:

Research Assistant
Faculty of Computer Science, HSE University
Borevskiy Andrey Olegovich

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Annotation

Besides the canonical right-handed B-form DNA, several other secondary DNA structures exist, including Z-DNA and G-quadruplexes (G4). Recent research has shown that these irregular DNA structures are associated with various diseases, such as diabetes, cancer, and Alzheimer’s disease. This makes the problem of predicting secondary structure regions in DNA sequences highly relevant.

Several general-purpose machine learning models have been developed for this task. However, most of these models rely solely on DNA sequence information for prediction. In particular, models for G4 overlook additional omics data, such as histone modifications, transcription factors, and other genomic features, which could potentially improve prediction accuracy. Some models for Z-DNA do incorporate omics data, but they are designed for general-purpose use, predicting secondary structures across the entire genome. Moreover, they show relatively low F1-score and, as a consequence, limited analytical ability.

In this work, we applied convolutional neural network models for prediction secondary DNA structures to more specific and complex dataset pluripotent stem cells. We conducted a series of experiments to identify the model with the highest prediction accuracy and used explainable AI (xAI) techniques to interpret the results, revealing key features that contribute to the model’s decisions.

Keywords

Bioinformatics, Z-DNA, G-quadruplexes, CNN, omics data, pluripotent stem cells, genomic sequence analysis, explainable AI (xAI)

1 Introduction

Following the discovery of the canonical right-handed B-form double helix of DNA (B-DNA) [1], other DNA secondary structures were also found to exist. One of them is a left-handed double-helical structure with a zigzag backbone, referred to as Z-DNA [2] (Figure 1.1). Another irregular structures are G-quadruplexes (G4s) [3], which are stable secondary structures of nucleic acids that occur when quartets of guanines are stabilized by a monovalent cation (Figure 1.2). These structures are transient, meaning that they are occasionally induced by biological activity and then revert back to canonical B-DNA form. An estimated frequency of Z-DNA and G4 occurrences among whole genome is about 0.3% – 1% and 1% – 2%, respectively [4, 5].

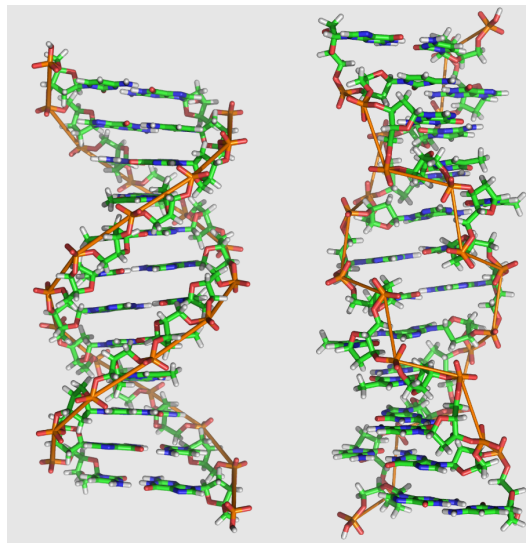


Figure 1.1: Different DNA forms: right-handed B-DNA (left) and left-handed Z-DNA (right).
Figure adapted with permission from [6] under a GNU Free Documentation license

Both secondary structures have been experimentally confirmed to be present in the human genome [8, 9]. Moreover, they are known to play important role in several biological mechanisms and are strictly connected with a variety of diseases.

For example, Z-DNA is associated with a higher probability of Alzheimer’s disease and cancer [10, 11]. As for G-quadruplexes, they are reported to participate in enhancer-promoter interactions, chromatin assembly, DNA replication, recombination, and repair [12]. They are also connected to various neurodegenerative diseases, such as fragile X syndrome, amyotrophic lateral sclerosis, frontotemporal dementia, and others [13]. Therefore, the detection and prediction of irregular DNA structures in genome sequences is a pressing modern problem, especially for the human genome, which is the focus of this paper.

Different methods have been proposed for predicting genome regions with Z-DNA and G-quadruplexes ranging from regular expressions and statistical approaches to complex machine

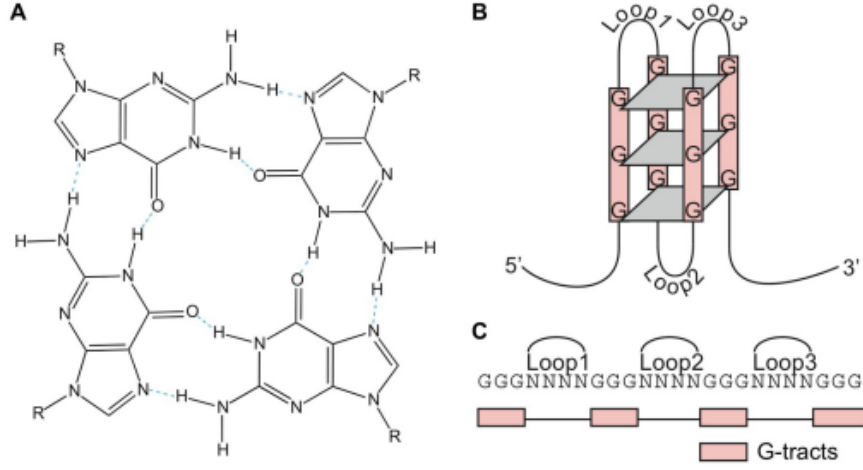


Figure 1.2: (A) Schema of a G-tetrad that makes up the core of a G4 structure. (B) Schematic representation of a parallel unimolecular G4. (C) Example of a single-stranded DNA G4 motif, consisting of four G-tracts, separated by three non-specific loops.

Figure adapted with permission from [7] under a CC BY license

learning models (we will discuss them later in Section 2). However, all of these approaches predict irregular structures across the whole genome, overlooking the potential importance of cell-specific predictions. Moreover, methods for G4 base their predictions solely on DNA sequence information, ignoring additional omics data such as transcription factors, histone modifications, and other features which may play not the least role in the formation of secondary structures.

In this work, we focus on pluripotent stem cells. Researching Z-DNA and G-quadruplexes in this case is crucial because such structures are involved in gene regulation and cellular differentiation processes that define stem cell behavior and their potential for regenerative medicine. As for architectures, we decided to stop on one of the most popular for genomic problems — convolutional neural networks. As a first approximation, we took a model derived from one of the latest CNN architectures developed by the HSE Bioinformatics Laboratory for Z-DNA prediction, which includes processing omics data. We trained this model specifically on pluripotent stem cells. We then used this model as a baseline for G-quadruplex prediction and conduct series of experiments in attempt to find the best architecture for G4 prediction. We also investigated different omics encodings to improve prediction accuracy for both types of secondary structures.

Another important question is which omics features are the most important for the formation of Z-DNA and G4 structures. Identifying such key features could be valuable for future biological research on the genomic functions of secondary structures. To address this, we analyzed the weights of the trained model and interpret them using an ensemble of explainable AI (xAI) methods. For Z-DNA, the most significant features have already been identified in general by HSE Bioinformatics laboratory, so we compared our results on pluripotent stem cells to highlight both

similarities and differences. For G4, no interpretation has been conducted so far, so we focused on identifying the key features. After interpretation, we retrained our models using a trimmed set of the most important features — adjusting its size to optimize accuracy — to validate the correctness of the interpretation. We also reviewed relevant biological articles to confirm the biological plausibility of the identified omics features involved in Z-DNA and G4 formation.

The rest of this paper is organized as follows. Section 2 reviews existing research and related solutions in the field. Section 3 describes the datasets, initial model architecture, evaluation metrics, and interpretation framework used for both types of secondary structures. Section 4 focuses on training a model for Z-DNA in pluripotent stem cells, identifying the most important omics features, comparing them with whole-genome data, and examining relevant biological literature. Section 5 presents experiments on model architecture for G-quadruplexes (G4s). Section 6 discusses the interpretation of omics features and relevant biological insights related to G4s. We conclude in Section 7.

All related code and notebooks are available on GitHub:

https://github.com/cutehater/ZDNA_G4_Research.

2 Literature Review

2.1 Z-DNA

One of the earliest successful attempts to predict Z-DNA-forming genomic regions was Z-Hunt [4] (and its subsequent improvement [14]) — a statistical approach based on mechanical and energetic considerations of the B- to Z-DNA transition, using only DNA sequence information. Z-Hunt remained the state-of-the-art solution for a long time.

In recent years, with the advancement of machine learning, deep learning approaches have been increasingly applied to this problem. One of them is DeepZ [15] — which explored recurrent neural networks (RNN), convolutional neural networks (CNN), and their combinations. In addition to DNA sequence, DeepZ incorporated various omics datasets — such as genome-wide maps of epigenetic modifications, transcription factor binding sites, RNA polymerase binding sites, and chromatin accessibility data. Another deep learning approach is GraphZ [16], which also uses both DNA sequence and omics data, but applies a different architecture — graph neural networks (GNN). However, both DeepZ and GraphZ demonstrate relatively low F1-scores, indicating their limited ability to handle class imbalance and revealing shortcomings in their analytical power for this task.

Moreover, cell- or tissue-specific aspects have not been studied extensively in this context. Considering that Z-DNA formation may vary between cell types, applying deep learning models to a more specialized context could yield new insights and enhance prediction accuracy. Therefore, in this work, we aimed to adapt a model architecture derived from one of the latest CNN models developed by the HSE Bioinformatics Laboratory to pluripotent stem cells and to interpret the results.

2.2 G-quadruplexes

Several experimental techniques have been developed for measuring and predicting G-quadruplex formation since their discovery. The earliest approaches were based on regular expression matching, assuming that specific sequence motifs could potentially form G4s [17]. Later, these methods were expanded and improved by taking more specific aspects into account. The most accurate and still widely used methods are G4Hunter [18], which considers G-richness and G-skewness, and Pqsfinder [19], which allows customization for detecting noncanonical G4s. However, all of these methods rely primarily on domain knowledge rather than direct experimental data, and their predictive power could be improved.

The second generation of methods applied to G-quadruplex prediction are machine learning approaches, achieving significantly better results. Over the years, a variety of architectures have been proposed for this problem. Early approaches such as Gboost [20] used gradient boosting models. These were later followed by neural network-based methods, including CNN and RNN architectures: G4Detector [21], PENGUINN [22], G4Mismatch [23]. More recently, large language models such as G-DNABERT have been introduced [24].

Nevertheless, all of these models are trained solely on DNA nucleotide sequences, missing the potential impact of omics data. This remains an open area for experiments, as incorporating omics data could further enhance prediction performance for G-quadruplexes. Another unexplored sphere is the biological meaning of omics features in the formation of G4 structures specifically in pluripotent stem cells. One of the main goals of this work was to interpret the weights of the trained models in order to explore this role.

2.3 Models interpretation

While modern deep learning models demonstrate high predictive performance across biological problems, their black-box nature often limits their direct applicability in scientific discovery. In particular, for tasks like predicting the formation of irregular secondary DNA structures, un-

derstanding which features affect the model’s prediction the most can be as important as the prediction itself. Model interpretability is crucial for two main reasons: firstly, to ensure the reliability of the results; and secondly, to potentially offer new biological insights by revealing sequence patterns, regulatory elements, or omics features associated with the formation of Z-DNA and G-quadruplexes.

Several techniques have been proposed for interpreting neural network decisions. Guided Backpropagation [25] and Deconvolution [26] compute the gradient of the target output with respect to the input, but backpropagation of activation functions is overridden so that only non-negative gradients are backpropagated. Both approaches were originally designed for the interpretation of convolutional neural networks and still are mainly used for CNN architectures. Integrated Gradients [27] calculates the integral of gradients with respect to inputs along the path from a given baseline to input. InputXGradient [28] takes the gradients of the output with respect to the input and multiplying by the input feature values. We discuss all these methods in more detail in Section 3.4.

In this work, we used an ensemble of state-of-the-art explainable AI (xAI) methods that complement each other. Combining these methods allowed us to smooth out the biases of individual approaches and obtain a more robust and comprehensive explanation of model behavior.

3 Setup and Environment

All computations and model training were performed on the HSE University Supercomputer [29].

3.1 Datasets for training and evaluation

For our problem, we need three key types of data: DNA sequences, omics data, and Z-DNA/G4 targets.

The DNA sequence is represented as a one-hot encoded matrix for the four nucleotides: adenine (A), cytosine (C), guanine (G), and thymine (T), organized by chromosomes. We used the latest human reference genome, hg38, which was obtained from the open-source UCSC Genome Browser [30].

The omics data represents various genomic annotations. For this study, we focus on omics data specific to pluripotent stem cells, which we retrieved from ChIP-Atlas [31] — a free public database and analysis platform that automatically collects, processes, and integrates raw sequenc-

ing data from a wide range of publicly available experiments. We incorporated six different categories of assays, which you can see in Table 3.1. We did not include the Unclassified and Input_Control categories, as they are irrelevant.

Table 3.1: Overview of assay categories in ChIP-Atlas.

Category	What it is	What it measures / detects
ATAC-seq	Assay for Transposase-Accessible Chromatin	Open chromatin regions
Bisulfite-seq	Whole-genome bisulfite sequencing	DNA methylation
DNase-seq	DNase I hypersensitive site sequencing	Open chromatin, hypersensitive regions
Histone	ChIP-seq for histone modifications	Chromatin state (active/silent, enhancer/promoter)
RNA_polymerase	ChIP-seq for RNA polymerase	Sites of active transcription
TFs_and_others	ChIP-seq for transcription factors and other chromatin-associated proteins	Transcription factor binding sites, co-factors, etc.

Each omics dataset is represented by genomic intervals (regions on chromosomes) where the signal is detected, along with a confidence metric expressed as $-10 \log(\text{MACS2 Q-value})$ (hereafter referred to simply as the Q-value). The Q-value is a positive number ranging from 1 to several thousand, depending on the dataset (with a maximum value of 5382 in our case). Lower values typically indicate noise, while higher values reflect stronger evidence for the presence of an omics signal in a given region.

For this work, we considered only omics intervals with Q-values of 50 or higher, as lower values are more likely to represent noise and can degrade model performance. Each omics was handled as separate feature. For each interval, we took the $\min(\text{Q-value}, 1000)$ and scale it to the range $[0, 1]$ (alternative omics encoding schemes have been explored in further experiments). The overall input data structure is illustrated in Figure 3.1.

The Z-DNA and G4 targets are represented as binary sequences, indicating whether a specific genomic region contains the corresponding secondary structure. Z-DNA targets were sourced from the Kouzine dataset [32], while G4 targets were obtained from the Endoquad database [33]. The Kouzine dataset is derived from a combination of experimental and computational approaches: single-stranded DNA regions identified experimentally were intersected with predictions made using Z-Hunt. We used all available intervals from this dataset. Endoquad is a comprehensive database that aggregates results from a variety of experimental studies designed to detect G-quadruplexes. Each identified G4 region is assigned a confidence level from 1 to 6, reflecting the number of independent experiments in which the structure was detected. In this study, we

Feature	1	2	3	4	5	6	7	8	9	10
A	1	0	0	0	1	0	0	0	1	0
G	0	1	0	0	0	1	0	1	0	1
C	0	0	1	0	0	0	1	0	0	0
T	0	0	0	1	0	0	0	0	0	0
ATAC-seq	0.91	1.00	0.55	0.77	0.22	0.48	0.18	0.88	0.96	0.54
DNase-seq	0.87	0.96	0.55	0.77	0.16	0.47	0.13	0.91	1.00	0.51
H3K4me3	0.83	0.41	0.09	0.08	1.00	0.68	0.02	0.34	0.88	0.36
H3K27ac	0.93	0.44	0.33	0.75	0.08	0.31	0.05	0.84	1.00	0.47
H3K4me1	0.54	0.94	1.00	0.46	0.17	0.96	0.43	0.13	0.60	0.85
H3K9me3	0.07	0.13	0.04	0.41	0.09	0.16	1.00	0.45	0.03	0.95
H3K27me3	0.11	0.07	0.09	0.06	0.42	0.37	1.00	0.13	0.07	0.88

Figure 3.1: Input data layout: DNA sequence concatenated with omics features.

considered only regions with a confidence level of 4 or higher.

For omics data collection and for converting all datasets into a sparse format (to enable compact storage and efficient retrieval), we used assistive frameworks developed by the HSE Bioinformatics Laboratory [34, 35] with several modifications.

Since Z-DNA and G4 occurrences are relatively rare across the entire genome (approximately 0.3–1% for Z-DNA and 1–2% for G4 [4, 5]), we balanced the training data to ensure effective model learning. Specifically, we selected all intervals containing at least one nucleotide associated with a Z-DNA or G4 occurrence and mixed them with randomly selected negative intervals (regions without the target structure) in a 1:3 ratio. We split the data into training and test sets using a 1:4 ratio, stratifying by the chromosome of each input interval. The initial genomic window size was set to 100; this parameter was also varied in subsequent experiments.

3.2 Initial model architecture

The initial convolutional neural network (CNN) architecture used for both secondary structure prediction tasks is shown in Figure 3.2. This model is derived from one of the latest CNN architectures developed by the HSE Bioinformatics Laboratory for the genome-wide identification of Z-DNA regions.

In this study, we retrained the model using data from pluripotent stem cells for subsequent interpretation for the Z-DNA prediction task. We also took the same architecture as a baseline for the G-quadruplex prediction task.

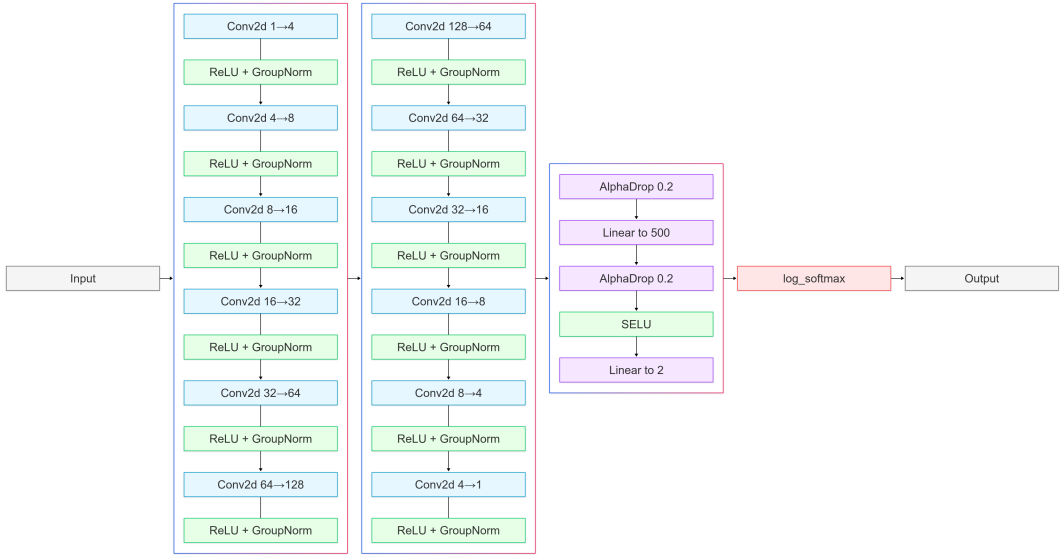


Figure 3.2: Initial CNN model architecture.

3.3 Metrics

The primary evaluation metric for our models was the F1-score, as it provides a balanced measure between precision and recall, which is particularly important in the context of high class imbalance present in our dataset. Relying only on accuracy in such scenarios can lead to misleading conclusions, since a model may achieve high accuracy by simply predicting the majority class. The F1-score, on the other hand, reflects the model's ability to correctly identify both positive and negative cases, offering a more reliable assessment under imbalanced conditions.

In addition to the F1-score, we evaluated model performance using the area under the receiver operating characteristic curve (ROC-AUC), which measures the trade-off between sensitivity and specificity across various classification thresholds. This metric helped ensure a comprehensive evaluation of the model's behavior.

3.4 Interpretation framework

As mentioned above, we chose four different interpretation techniques: Integrated Gradients [27], InputXGradient [28], Deconvolution [26], and Guided Backpropagation [25]. These methods are quite different, each having its own advantages, limitations, and inherent biases. Therefore, it is reasonable to combine their results to smooth out individual weaknesses and obtain a more balanced and reliable estimate of feature importance.

Integrated Gradients is a path-based method that attributes input feature importance by integrating the gradients of the model's output with respect to the input along a straight-line path from a baseline (typically a zero vector) to the actual input. The attribution for the i -th feature

is given by:

$$\text{IG}_i(x) = (x_i - x'_i) \int_{\alpha=0}^1 \frac{\partial f(x' + \alpha(x - x'))}{\partial x_i} d\alpha,$$

where x is the input, x' is the baseline, and f is the model. Thus, IG represents the mean impact of each feature on the prediction of the model.

InputXGradient is a simpler technique that multiplies the input feature by the gradient of the models output with respect to that feature::

$$\text{IXG}_i(x) = x_i \cdot \frac{\partial f(x)}{\partial x_i}.$$

It can be interpreted as a linear model: the gradients are coefficients of each input. The product of the input with a coefficient refers to the total contribution of the feature to the models output.

Deconvolution is a gradient-based visualization method originally proposed for convolutional neural networks (CNNs). Unlike standard backpropagation, it modifies the gradient flow by using only the positive parts of the gradient during the backward pass, suppressing negative influences.

Guided Backpropagation refines Deconvolution by combining it with standard backpropagation. It modifies both the forward ReLU and the backward gradient by masking out negative values in the backward pass.

In this study, we used HSE Bioinformatics Laboratory framework [36] (with a number of modifications) which integrates all four interpretation methods. For each method, the framework computes the mean attribution across all true positive samples from both the training and test datasets. We implemented two types of true positive clipping: prefiltering, which clips before attribution by wrapping the model with a custom forward pass (applicable only to Integrated Gradients and InputXGradient), and postfiltering, which clips after attribution by directly filtering true positives (compatible with all four methods). We experimented with different combinations of these approaches to achieve the most robust and meaningful interpretation. Then, for each feature, it calculates the mean absolute deviation percentage relative to the other features for every method. Finally, it averages these deviations across all used methods to obtain a single consolidated importance score for each feature. Features with the highest mean deviation are considered the most significant for the models predictions. Using this ensemble of xAI methods allows us to produce a more robust and trustworthy interpretation of the model’s behavior.

4 Z-DNA Models and Interpretation

4.1 Model training

For Z-DNA prediction, we used the model described in Subsection 3.2. The original model was trained on the entire genome for 20 epochs, without dividing the data by cell types or tissues. We re-trained this model specifically on data from pluripotent stem cells, experimenting with various omics encodings: (1) $\min(\text{Q-value}, 1000)/1000$, (2) binary encoding (1 for each interval present in the omics data, 0 otherwise), (3) Q-value scaled by the maximum value across all omics features, (4) Q-value of each omics feature is scaled by dividing by the maximum value of this feature, and (5) Q-value of each omics feature is scaled by subtracting mean of this feature and dividing by its variance. The results are presented in Table 4.1.

Table 4.1: Z-DNA model performance with different omics encodings.

		Classification metric			
		F1-score	ROC-AUC	Precision	Recall
General case		88.15%	97.71%	89.41%	87.09%
Stem cells	Scaled by 1000	87.53%	97.94%	91.70%	83.93%
	Binary	83.66%	97.60%	91.10%	77.63%
	Scaled by overall max	88.11%	97.78%	91.35%	85.27%
	Scaled by individual max	87.01%	97.89%	91.46%	83.19%
	Scaled by individual mean/var	85.01%	97.56%	91.68%	79.47%

As we can see, the overall max scaled omics encoding yielded the highest F1-score, close to general case, so we used this version for subsequent interpretation.

4.2 Omics interpretation

We used the best-performing model for Z-DNA prediction and interpreted it using our framework. Deconvolution method turned out to be mostly noisy, as it mixed the importance of nucleotids and omics features, so we excluded it from the final ranking. For the remaining three methods, we applied both postfiltering (for all three methods) and prefiltering (for Integrated Gradients and InputXGradient). The rankings from both filtering types showed similar results. Nucleotides, as expected, turned out to be the most important features by a wide margin (Table 4.2).

We experimented with the number of omics features to keep in order to optimize the predictive performance of the best model. For these experiments, we used the prefiltering interpretation approach. Once we identified the optimal number of omics features, we also tested postfiltering.

Table 4.2: Mean deviation of nucleotids for Z-DNA in stem cells (ranked by Integrated Gradients, InputXGradient and Guided Backpropagation, postfiltering).

Feature	Mean percentage deviation
C	33136%
G	30155%
T	27157%
A	14978%
The most important omics	320%

The best F1-score was achieved by postfiltering with Integrated Gradients and InputXGradient methods and 50 omics features. The results are shown on Table 4.3.

Table 4.3: Best Z-DNA model performance with varying numbers of top omics features kept.

Filter type	Top omics features count	Classification metric			
		F1-score	ROC-AUC	Precision	Recall
Prefiltering (IG+IXG)	10	89.77%	95.75%	89.66%	89.97%
	20	89.92%	95.68%	89.34%	90.58%
	30	89.85%	95.75%	89.56%	90.23%
	50	89.93%	95.90%	91.26%	88.72%
	100	89.89%	95.96%	91.11%	88.78%
Postfiltering (IG+IXG)	50	89.96%	95.96%	90.80%	89.21%
Postfiltering (IG+IXG+GB)	50	89.79%	95.89%	90.49%	89.18%

The ten most important omics features identified through the interpretation are listed in Table 4.4.

Table 4.4: Top 10 omics features contributing to Z-DNA prediction in stem cells. Each omics feature is followed by its rank in parentheses.

Omics category	Features (rank)
Histone modifications	H3K4me3 (1), H2A.Z (5), H4K20me1 (10)
Transcription factors	BRCA1 (2), ZMAT4 (4), KDM2A (6), EPOP (7), ISL1 (8), CHD6 (9)
RNA polymerase	RNA polymerase II (3)

4.3 Biological relevance

We tried to analyze the most important omics features identified by our interpretation and to link them to findings in relevant biological literature.

The most important one, H3K4me3, is indeed highly associated with Z-DNA formation according to biological studies. It is an active histone mark enriched in the "Z compartment"

of pluripotent stem cells, which is associated with Z-DNA formation and totipotent-like states. During the pluripotent-to-totipotent transition, a significant decrease in H3K4me3 levels was observed alongside Z-DNA dynamics, suggesting a close relationship between this mark and Z-DNA-mediated chromatin states [37].

KDM2A regulates germ cell genes and endogenous retroviruses in embryonic stem cells by demethylating H3K36me2 and influencing H3K4me3 deposition [38]. Given H3K4me3s link to Z-DNA regions, KDM2A may indirectly affect Z-DNA formation through its epigenetic regulation of chromatin marks.

For other omics features, we did not find direct evidence in the literature linking them to Z-DNA formation. However, their roles may not have been extensively studied in this context, and they may influence or be influenced by Z-DNA formation indirectly through their active roles in transcription and chromatin dynamics. For example, RNA Polymerase II (RNA Pol II) activity is linked to transcriptional regulation during cell fate transitions in pluripotent cells. Cell-specific chromatin loops involving RNA Pol II elongation are important for gene expression changes, which may indirectly relate to chromatin structure states including Z-DNA [39]. Similarly, H2A.Z is critical for establishing bivalent chromatin domains in pluripotent stem cells and is involved in chromatin remodeling and gene repression [40]. It might influence chromatin states permissive for Z-DNA since its formation correlates with chromatin accessibility and remodeling.

4.4 Comparison to whole genome case

We compared the top 10 most important omics features identified by our model for pluripotent stem cells with those from a general-purpose model by the HSE Bioinformatics Laboratory. The general model was trained on the hg19 genome and used a ranking based on the mean absolute percentage deviation across all four interpretation methods: Integrated Gradients, InputX-Gradient, Deconvolution, and Guided Backpropagation. The comparison results are presented in Table 4.5.

As we see, several omics features are important in both models, including H3K4me3, RNA Polymerase II, H3K27me3, and H2A.Z. However, there are notable differences. For instance, H3K4me2, which ranks first in the general case, is not significant for stem cells. This may be due to lower expression or occurrence of this histone modification in pluripotent stem cells compared to the whole genome. Additionally, differences in transcription factors suggest a cell-type-specific regulatory landscape, highlighting distinctions between stem cells and the general genomic context.

Table 4.5: Comparison of the top omics features for Z-DNA in pluripotent stem cells and the general case. NP indicates that the omics feature is not present in stem cells.

Rank	General case top		Stem cells top	
	Omics feature	Other top rank	Omics feature	Other top rank
1	H3K4me2	198	H3K4me3	2
2	H3K4me3	1	BRCA1	423
3	ING2	NP	RNA Polymerase II	5
4	NPM1	NP	ZMAT4	803
5	RNA Polymerase II	3	H2A.Z	23
6	CTCF	321	KDM2A	350
7	H3K4me1	248	EPOP	1719
8	H3K27ac	255	ISL1	840
9	H3K27me3	15	CHD6	151
10	Epitope tags	320	H4K20me1	406

5 G4 Models Experiments

5.1 Baseline results

First, we trained the initial model (see 3.2) on the G-quadruplex target. The model achieved relatively strong performance with F1-score about 68%. However, we conducted further experiments in an effort to improve this result. The model mentioned above was trained on 20 epochs, as the original one. Meanwhile, for the subsequent models discussed in this section, we trained them for 812 epochs. This choice was driven by two factors:

- The amount of data available for G4 is significantly larger than for Z-DNA.
- The models we experimented with contain considerably more parameters.

As a result, training these models is very time-consuming, even with extensive GPU parallelization. Additionally, we observed that model performance typically reaches a plateau in the final epochs even when using a learning rate scheduler making further training ineffective. Additionally, several experiments were terminated in early epochs if the F1-score showed noticeably lower performance relative to the current best model.

5.2 Optuning high-level model parameters

As a next step, we focused on selecting the optimal high-level model architecture using Optuna [41] — an automated hyperparameter optimization framework. The core design remained consistent: the model consists of two main parts, with the first half comprising convolutional layers

with an increasing number of channels for pattern extraction, and the second half comprising convolutional layers with a decreasing number of channels for compressing relevant information.

We tuned a wide range of parameters, including the number of convolutional layers, kernel sizes, dilation and stride settings, the number of channels at each layer (by selecting a multiplier or divisor as powers of two), as well as capping the maximum number of channels at 512. In addition, we explored different configurations for the linear layers, optimizer choices, and other training-related parameters. The full list of parameters and their respective search intervals is provided in Table 5.1.

Each model was trained for 4 epochs without any learning rate scheduling. For early stopping and pruning, we applied MedianPruner and also terminated trials that achieved an F1-score of 0 after the first epoch. For trial sampling, we used TPESampler. We also applied Batch Normalization to all convolutional layers, as it is widely considered one of the most effective normalization techniques for CNNs and does not require additional tuning (unlike GroupNorm, which needs the number of groups to be specified).

Table 5.1: Hyperparameter search space used for high-level Optuna tuning.

Parameter	Type / Distribution	Range / Choices
Activation layer type	Categorical	{ReLU, LeakyReLU, PReLU, ELU, SELU}
Number of convolutional layers	Integer	2 to 16 (step 2)
Kernel size	Categorical	{2, 3, 5, 7}
Channels number mult./div. (each layer)	Categorical	{2, 4, 8}
Stride (each layer)	Categorical	{1, 2}
Dilation (each layer)	Categorical	{1, 2}
Linear layer size	LogUniform Integer	$\text{width} \times 4$ to $\text{width} \times 40$
Dropout before linear layer	Float (Step 0.05)	0.15 to 0.75
Dropout after linear layer	Float (Step 0.05)	0.15 to 0.75
Optimizer type	Categorical	{Adam, RMSprop, SGD, AdamW}
Learning rate	LogUniform Float	10^{-5} to 10^{-3}
Weight decay	LogUniform Float	10^{-6} to 10^{-4}
Momentum for SGD	Uniform Float	0.9 to 0.99

In total, we conducted 25 optimization trials. The best architecture discovered consisted of 14 layers and achieved an F1-score of 70.4% after 4 epochs. However, we tested a reduced version of the model with 10 layers and observed that the difference in F1-score was negligible, while the training time was significantly reduced. Therefore, we selected the 10-layer model for further optimization. Its architecture is shown in Figure 5.1. The best optimizer turned out to be AdamW with the learning rate about $2 \cdot 10^{-4}$ and weight decay about $5 \cdot 10^{-5}$.

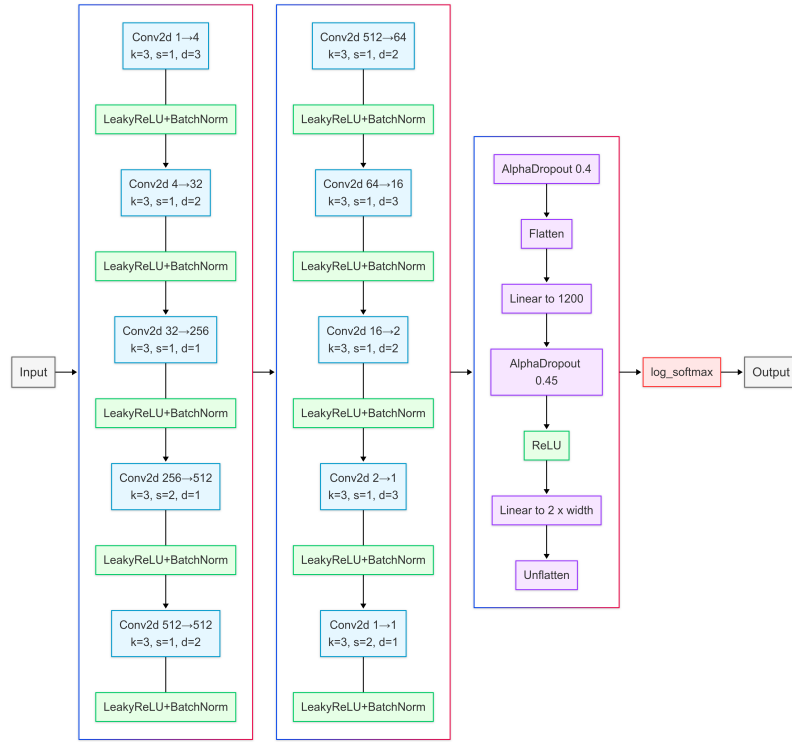


Figure 5.1: First approximation of the model architecture for G4 selected by optuna.

5.3 Training the tuned model for more epochs

First, we attempted to train the optimal model found by Optuna for a larger number of epochs. However, the performance plateaued after the second epoch. The best F1-score remained unchanged.

5.4 Removing dilations and using SELU activation function

We attempted to remove dilations, as they significantly increase computation time. Additionally, we replaced the activation function in the fully connected layer with SELU, which is widely known to perform well in combination with AlphaDropout. However, training was stopped after the first epoch, as the model exhibited poor performance (F1-score about 41%). These results suggest that dilations and consequently the ability to capture long-range dependencies between sparse nucleotides are crucial for achieving good model performance.

5.5 Adding a scheduler and removing the final stride

After selecting the initial model parameters using Optuna, we introduced a learning rate scheduler in an attempt to further improve performance, and trained the model for 10 epochs. We used the `ReduceLROnPlateau` scheduler with the following parameters: *patience* = 0 and *factor* = 0.2. In addition, we removed the stride from the last convolutional layer, as it introduced

unnecessary downsampling given the small number of parameters at this stage. The model was trained for 10 epochs and showed a 0.6% increase in F1-score.

5.6 Adding scheduler and residual connections

We reduced the number of convolutional layers by two and added residual connections. Each residual block applies an identity mapping to the input and performs element-wise addition: $F(input) + input$, using a 1×1 convolution to match dimensions where necessary. This design is widely known to improve gradient flow and convergence. We also introduced a learning rate scheduler in an attempt to further improve performance. We used the `ReduceLROnPlateau` scheduler with the following parameters: *patience* = 0 and *factor* = 0.2. As a result, the F1-score improved to 72.1% after 12 training epochs.

5.7 Adding dropout after convolutional layers

We tried adding dropout not only in the fully connected layers, but also between the convolutional layers. We also replaced AlphaDropout with standard Dropout for the fully connected layers (with the same drop probability). We used a dropout probability $p = 0.1$. However, the F1-score after 5 epochs showed no improvement.

5.8 Reducing the number of convolutional channels

We reduced the maximum number of channels from this sequence:

$$[1 \rightarrow 4 \rightarrow 32 \rightarrow 256 \rightarrow 512 \rightarrow 128 \rightarrow 16 \rightarrow 4 \rightarrow 1]$$

To the following one:

$$[1 \rightarrow 4 \rightarrow 16 \rightarrow 64 \rightarrow 128 \rightarrow 64 \rightarrow 16 \rightarrow 4 \rightarrow 1]$$

We also experimented with the fully connected layer: increasing its size from 1200 to 2200, decreasing the dropout probability before it from 0.4 to 0.1, and after it from 0.45 to 0.2. The F1-score decreased by approximately 1% compared to the best model. However, this model trains about 40% faster, so we used it for width experiments (with later retraining larger model with the best parameters).

5.9 Interval width experiments

We experimented with varying the length of the nucleotide interval. Starting from an initial length of 100, we also tested lengths of 200 and 300, but observed that prediction quality decreased as the interval length increased. This prompted us to explore shorter intervals 75, 50, and 30 nucleotides. The best performance was achieved with a length of 50, reaching an F1-score of approximately 72.1%, so we continued with this value.

5.10 Omics encoding experiments

We experimented with various omics encodings besides the initial scheme ($\min(\text{Q-value}, 1000)/1000$), trying the same four encoding types as in the Z-DNA case (Section 4.1). Binarizing the omics data resulted in significantly worse performance, with an F1-score of only 51.6% after 2 epochs, so training was stopped. Among the remaining three, the best result — an F1-score of approximately 73.1% after 10 epochs — was again achieved using the overall max-scaled omics encoding.

We then trained a larger model (with an increased number of channels) using the optimal interval length and omics encoding. This led to a slight F1-score improvement of 0.1% after 15 epochs.

5.11 Results summary

Table 5.2 summarizes the metrics for all conducted experiments.

The final model architecture is shown on Figure 5.2.

6 G4 Interpretation

6.1 Omics interpretation

We used the best-performing model for G4 prediction and interpreted it using our framework. Deconvolution and Guided Backpropagation methods were found to be mostly noisy, as they mixed the importance of nucleotids and omics features and produced outlier-level percentage deviations, so we excluded them from the final ranking. For the remaining two methods, Integrated Gradients and InputXGradient, we applied both postfiltering and prefiltering. We also interpreted the best models trained on two interval widths: the initial 100 and the later-optimized 50. The resulting rankings were consistent across both filtering types. Nucleotides again turned out to be

Table 5.2: Summary of experimental results across various model configurations.

Category	Experiment	Classification metric			
		F1-score	ROC-AUC	Precision	Recall
	Baseline	67.86%	85.10%	70.93%	65.31%
	Optuna best model	70.43%	86.15%	66.88%	74.68%
	Remove dilations + SELU	41.20%	80.22%	30.50%	41.20%
	Residuals, 8 conv layers, add scheduler	72.13%	88.80%	72.28%	72.27%
	Add dropout after conv layers	69.55%	86.64%	74.10%	65.82%
	Fewer conv channels	70.99%	88.51%	74.70%	67.93%
Interval experiments (reduced channels)	Length 200	66.19%	90.17%	69.84%	63.17%
	Length 300	60.52%	89.65%	65.21%	56.76%
	Length 75	70.89%	87.02%	76.51%	66.36%
	Length 50	72.14%	80.65%	72.95%	71.71%
	Length 30	71.23%	77.25%	77.06%	66.71%
Omics encoding experiments (reduced channels)	Binary	51.60%	79.16%	75.30%	39.90%
	Scaled by overall max	73.14%	81.92%	73.40%	73.22%
	Scaled by individual max	71.57%	82.82%	76.59%	67.52%
	Scaled by individual mean/var	69.63%	79.04%	67.71%	69.64%
Big model with best parameters		73.25%	84.36%	75.40%	71.55%

the most important features by a wide margin (Table 6.1). However, the difference between the most important omics and the least important nucleotide is relatively less than for Z-DNA (5-fold vs. 50-fold), suggesting that omics data plays a more significant role in G4 detection.

Table 6.1: Mean percentage deviation of nucleotids for G4 in stem cells (ranked by Integrated Gradients and InputXGradient, prefiltering, interval width 50).

Feature	Mean percentage deviation
G	148379%
A	134705%
C	126391%
T	75949%
The most important omics	16885%

We experimented with the number of omics features to keep in order to optimize the predictive performance of the best model. For these experiments, we used the prefiltering interpretation approach and width 50. Once we identified the optimal number of omics features, we also tested postfiltering using the same width. The best F1-score was achieved by prefiltering and 30 omics features. The results are shown on Table 6.2

The top ten ranked omics for both filtering types and interval widths are listed in Table 6.3.

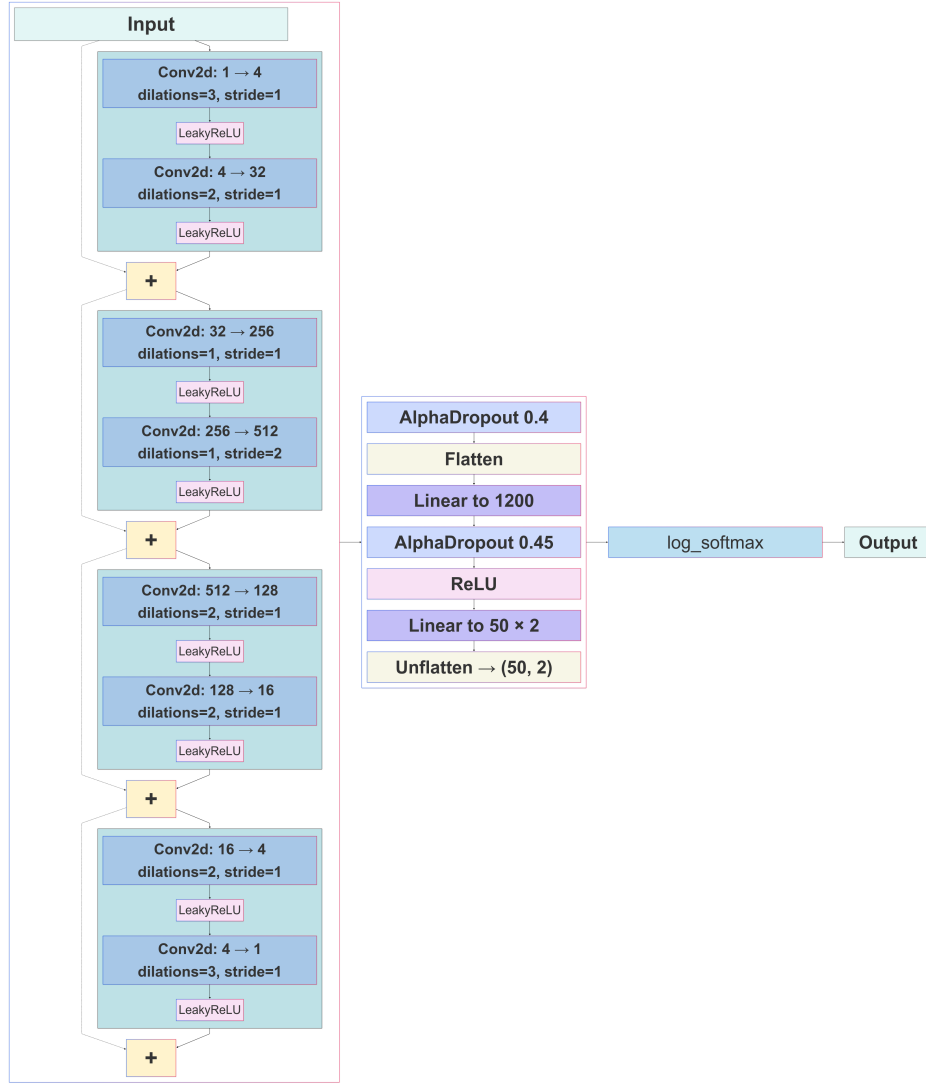


Figure 5.2: Final CNN model architecture for G4.

Table 6.2: Best G4 model performance with varying numbers of top omics features kept (ranked by Integrated Gradients and InputXGradient, interval width 50).

Filter type	Top omics features count	Classification metric			
		F1-score	ROC-AUC	Precision	Recall
Prefiltering	10	76.35%	83.42%	75.23%	77.75%
	20	76.59%	83.80%	75.45%	78.02%
	30	76.77%	83.65%	74.58%	79.35%
	40	76.40%	84.03%	76.42%	76.62%
	50	76.44%	83.84%	75.69%	77.43%
	100	76.40%	84.03%	76.42%	76.62%
Postfiltering	50	76.64%	83.59%	74.83%	78.78%

6.2 Biological relevance

As we can see from Table 6.3, across most ranking types, the consistently top-ranked omics features included transcription factors TP53BP1, BCOR, and KDM1A; histone modifications

Table 6.3: Top omics features for G4 prediction by filtering method and interval width (ranked by Integrated Gradients and InputXGradient).

Rank	Prefiltering		Postfiltering	
	Width 50	Width 100	Width 50	Width 100
1	TP53BP1	H3K4me3	BCOR	BCOR
2	KDM1A	KDM1A	ATAC-Seq	ATAC-Seq
3	BCOR	DNase-Seq	H3K27ac	H3K4me3
4	H3K4me3	TFEB	DNase-Seq	Epitope tags
5	NANOG	BCOR	CTCF	H3K27ac
6	ATAC-Seq	EP300	YY1	TP53BP1
7	TFEB	Epitope tags	KDM1A	DNase-Seq
8	CREB1	TP53BP1	H3K4me3	RNA Polymerase II
9	H3K27ac	ATAC-Seq	MAX	TFAP2A
10	KDM2B	H3K27ac	TP53BP1	H3K27me3

H3K4me3 and H3K27ac; as well as chromatin accessibility signals from ATAC-seq and DNase-seq. Additionally, the transcription factor TFEB appeared prominently under prefiltering. We tried to analyze these features and to link them to findings in relevant biological literature.

We found out that co-localized G4s and R-loops are enriched at active promoters marked by H3K4me3 and at active enhancers marked by H3K27ac. These epigenetic marks correlate with transcriptionally active regions where G4 structures are known to play regulatory roles [42].

Regarding ATAC-seq and DNase-seq, G4s frequently occur in accessible chromatin regions detected by these assays. Active promoters/enhancers (identified by ATAC/DNase) show co-occurrence with G4s and R-loops and enrichment of G-rich motifs at transcription start sites [42], as well as influence on DNA methylation patterns at CpG islands, which are tightly associated with G4 presence in human genome [43].

We did not find direct evidence linking G-quadruplexes to the transcription factors mentioned above. However, these connections may not yet be well-studied, or the connection may be indirect. For instance, KDM1A’s role in histone demethylation at enhancers overlaps spatially with G4/R-loop-enriched regulatory regions [44]. Similarly, studies have shown that unresolved G4s can cause replication fork stalling and DNA damage, which is recognized by DNA damage markers such as TP53BP1 [45]. This indicates that TP53BP1 is involved in responding to DNA damage caused by G4 structures.

7 Conclusion

In this study, we explored the application of convolutional neural networks for predicting secondary DNA structures specifically Z-DNA and G-quadruplexes (G4) in pluripotent stem cells,

incorporating both sequence and omics data. Our work built upon existing approaches while introducing several key innovations, including cell-specific modeling and the incorporation of omics data for G4 prediction – a novel aspect not previously explored in the literature.

For Z-DNA prediction, we adapted a CNN model originally designed for genome-wide analysis and fine-tuned it for pluripotent stem cells, achieving final F1-score of 89.96%. Through explainable AI techniques, we identified key omics features influencing Z-DNA formation, including H3K4me3, BRCA1, and RNA Polymerase II, and validated their biological relevance through existing literature. Notably, we observed differences in feature importance between stem cells and the general genome, highlighting the significance of cell-specific regulatory mechanisms.

For G-quadruplex prediction, we conducted extensive architectural optimization, including hyperparameter tuning, residual connections, interval width and omics encoding experiments. We achieved final F1-score of 76.77%, a significant improvement over baseline approaches. Interpretation of this model revealed that TP53BP1, H3K4me3, and chromatin accessibility signals (ATAC-seq/DNase-seq) were among the most influential omics features, suggesting a strong link between G4 formation and transcriptional regulation in stem cells.

Future work could explore more advanced architectures, investigate additional cell types, and experimentally validate the identified omics features to further research their roles in secondary DNA structure dynamics. Alongside omics features, incorporating k-mers (sequences of k consecutive nucleotids) analysis could improve prediction accuracy and provide deeper biological insights. By interpreting k-mers importance, we may uncover novel sequence motifs or structural patterns linked to Z-DNA and G4 formation, further bridging computational predictions with experimental biology.

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